



A two-stage expert-guided reinforcement learning controller training strategy for large time-delay systems

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ABSTRACT

The control of large time-delay systems presents a considerable challenge and practical necessity. Applying deep reinforcement learning (DRL) controllers for such systems encounters training difficulties due to the delayed rewards during the training process. Multi-stage training via incrementally increasing time delay is time-consuming and lacks robustness. To address these issues, we develop a two-stage expert-guided training framework for DRL controllers, where a Proportional-Integral-Derivative (PID) controller is adopted as an expert to accelerate the training procedure. A reference control variable trajectory generated by the PID controller and a hypothetical superior PID controller is utilized to calculate the rewards for the DRL controller during the stages of learning the expert and surpassing the expert, respectively. Two case studies demonstrate the availability and efficiency of the proposed strategy. For the industrial case study with a time delay of 264.8 s, the proposed strategy effectively trains DRL controllers, which perform better than PID controllers.

1. Introduction

Time delays, a common phenomenon in process control (Deepa and Sugumaran, 2011), typically refer to the latency between when a system's input changes and when its output responds. This delay can be caused by various factors, including physical transmission delays (e.g., fluid flow in pipes), the dynamic characteristics of chemical reactions, measurement delays from sensors, and action delays from actuators. The presence of these delays not only increases the complexity of designing control systems but also leads to a decline in system performance and even instability, especially when the length of the delay is close to or exceeds the system's dynamic response time. Exploring how to overcome the challenges posed by time delays has continuously been a hot topic of research.

To enhance the capability of PID control in addressing time delay issues (Abbasi et al., 2019; Gaing, 2004; George and Ganesan, 2022), traditional methods have incorporated various strategies, such as predictive control, delay compensation techniques, and gain scheduling methods. Predictive control methods, such as the Smith Predictor (Smith, 1959), forecast future process outputs to mitigate time delays

and have been enhanced by various researchers (Li et al., 2023; Liu and Wang, 2010; Pei et al., 2021; Veeramachaneni et al., 2013; Wang et al., 2023; Zhang et al., 2019). Delay compensation techniques use a rational function, where both the numerator and denominator are polynomials, to approximate complex target functions. First-order Pade approximation is one of the most popular approximation methods to represent time delay elements over the past two decades (Kun et al., 2015; Paul et al., 2013; Shamsuzzoha et al., 2009, 2014; Suryavanshi et al., 2019; Yang et al., 2008). Time delay systems can also be handled using a PID controller with gain scheduling, and several researchers have contributed to this aspect (Bolea et al., 2014; Hermassi et al., 2023; Kissoum et al., 2023). Gain Scheduling is based on predefined rules or models. The fundamental idea is that the parameters of the control system are dynamically adjusted in response to changes in the system's operating point or certain key parameters. It should be noted that the above strategies based on predictive control, delay compensation techniques, and gain scheduling methods rely on accurate knowledge of the process model (Chowdhury and Lu, 2022; Chowdhury et al., 2023; George and Ganesan, 2022). Adjusting and optimizing control strategies remain challenging when faced with complex or changing system dynamics.

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The reward-based reinforcement learning technique has demonstrated strong performance and promising application prospects in process control (Del Real Torres et al., 2022; Faria et al., 2022; Jhaveri et al., 2022). Similarly, deep learning has shown substantial potential (Hansen et al., 2023; Okamoto et al., 2024; Zhang et al., 2023b). Reinforcement learning when combined with deep learning, becomes deep reinforcement learning. The application of reinforcement learning in various fields, such as power systems (Chen et al., 2024; Zhang et al., 2024), navigation (Chung et al., 2024; Tang et al., 2023; Xiao et al., 2024), new energy vehicles (He et al., 2024; Wu et al., 2024), and robotics (Cai et al., 2023; Huang et al., 2024; Peng et al., 2024), has been reported. In our previous work (Hong et al., 2024), we developed a deep reinforcement learning controller for the temperature control of the polyethylene fluidized bed reactor. The proposed controllers based on the Deep Deterministic Policy Gradient (DDPG) algorithms have demonstrated superior performances than traditional PID controllers in case studies.

Reinforcement learning can also be applied to control problems with time delays. As early as 2003, researchers (Katsikopoulos and Engelbrecht, 2003) began to explore overcoming Markov decision processes (MDP) with time delays. They expanded the state space method to transform the problem into a standard MDP without time delays and extended the MDP to consider both current and past states to incorporate time delays. However, these approaches introduce increased computational complexity, which may surpass the challenges presented by the delays themselves. Researchers have also incorporated conventional methods to address time delay issues in reinforcement learning control, such as first-order Pade approximation methods (Zhuang et al., 2018) and PID-based errors (Chen et al., 2022). Zhuang et al. (2018) used a first-order Pade approximation method to address the time delay (1 s) in a reinforcement learning controller for the pulp and paper industry. Chen et al. (2022) utilized an improved TD3 algorithm (ITD3) to overcome time delay issues in the Selective Catalytic Reduction (SCR) denitrification system (13 s), where the algorithm performed differential and integral operations on the error between the set value and the measured value. Besides, model predictive control has also been applied, considering its predictive abilities. Xie et al. (2020) adopted a model predictive control-guided reinforcement learning control scheme (MP-RLC) to overcome a time delay of 7 s. Deng et al. (2021) combined model predictive control with reinforcement learning in turbidity control of sedimentation tanks in water treatment plant control systems, whose time delay is 10 s. Siraskar (2021) employed a multi-stage training method, progressively introducing control-challenging factors such as friction and time delays, to address time delay (2.5 s) in a valve control case. Researchers also introduced new concepts into reinforcement learning, such as the M network (Liang et al., 2018), new state variables (J. Zhang et al., 2023a) and delay-based two-dimensional Bellman equation (Shi et al., 2024). Liang et al. (2018) introduced the concept of a delay matrix function network (M network) in reinforcement learning, utilizing the M network to approximate time-delay control signals (0.15 and 1 s). J. Zhang et al. (2023a) introduced new state variables to account for both the current state of the intelligent agent and the impact of historical control inputs caused by input delays on the state, transforming system equations with delays into no-delay forms. Random time delay of 0–1 s were addressed. Shi et al. (2024) developed a delay-based two-dimensional Bellman equation to address time delay issues, focusing on mathematical derivations without specifying actual time delay values.

A summary of literature using PID and reinforcement learning for control systems with time delay is shown in Table 1. The time delay ranges from 0.07 s to 180 s. It can be observed that the time delays addressed by traditional solutions (such as PID+Smith) are significantly broader, up to 180 s. While PID controllers remain effective for many time-delay processes, their performance can deteriorate when facing highly nonlinear, time-varying, or large-delay systems. In contrast, reinforcement learning controllers offer enhanced adaptability and

Table 1

A summary table of the time delay information.

author	time delay, seconds	Problem solved	Resolution
(Liu and Wang, 2010)	10	time delay systems	PID+Smith predictor
(Veeramachaneni et al., 2013)	0.6	closed-loop system with a time delay	PID+Smith predictor
(Zhang et al., 2019)	180	chemical dosing system in power plants	PID+Smith Predictor
(Pei et al., 2021)	0.07	A compliant constant-force actuator based on the cylinder	PID+Smith predictor
(C. Li et al., 2023)	6.57	cascade processes with a time delay	PID+Smith predictor
(Wang et al., 2023)	10	fertilization systems	PID+Smith predictor
(Yang et al., 2008)	16.27	stable first order and high order systems with time delay.	Pade approximation
(Shamsuzzoha et al., 2009)	6	various types of time-delay process	Pade approximation
(Paul et al., 2013)	7.4	pure integrating process with delay (IPD)	Pade approximation
(Paul et al., 2014)	7.5	purely integrating processes with dead time	Pade approximation
(Kun et al., 2015)	10	industrial boiler combustion control	Pade approximation
(Suryavanshi et al., 2019)	7.4	time delay process with stable and integrating process type model	Pade approximation
(Bolea et al., 2014)	61	Lunax Galleri flow regulating	Gain Scheduled
(Hermassi et al., 2023)	–	grid-connected wind energy conversion system	Gain Scheduled
(Kissoum et al., 2023)	8	fractional order linear parameter varying systems with time-varying delay	Gain Scheduled
(Zhuang et al., 2018)	1	A Markov decision process of basis weight and moisture control	Reinforcement Learning
(Liang et al., 2018)	1	unknown nonlinear discrete-time systems with time delays	Reinforcement Learning
(Xie et al., 2020)	7	third-order linear system and nonlinear CSTR system with a large time delay	Reinforcement Learning
(J. Zhang et al., 2023a)	0–1	multiagent systems with input delay	Reinforcement Learning
(Siraskar, 2021)	2.5	Reinforcement learning for control of valves	Reinforcement Learning
(Deng et al., 2021)	10	The water treatment system for water purification plant	Reinforcement Learning
(Chen et al., 2022)	13	selective catalytic reduction denitration system	Reinforcement Learning

optimization capability, learning control policies that are difficult to achieve through traditional PID tuning alone. As shown in Table 1, the time delays addressed by reinforcement learning solutions are somewhat narrower, with the maximum only reaching 13 s. However, for the process industry, time delay issues may reach hundreds of seconds. Addressing large time delay issues of several hundred seconds in the process industry necessitates the development of new reinforcement learning methods.

Training reinforcement learning controllers for large time-delay systems is challenging. The effect of the action is delayed and cannot be easily observed, leading to mismatched reward-state pairs. Thus, quantifying the performance or the reward for the action is difficult. Directly employing standard reinforcement learning is often inefficient and needs an excessive number of training iterations due to low sample

efficiency (Gupta and Najjaran, 2022). Multi-stage learning via incrementally increasing time delay might be feasible, as the controller will gradually learn the system’s dynamic features. However, it could be time-consuming and might fail once any training step encounters difficulties. Model predictive based methods may also encounter the issues of low predictive accuracy for large-delay systems. Learning from a demonstration, who has already known some system features, can be an efficient method. It leads to Learning from Demonstrations (LfD), which is a machine learning strategy that acquires complex task executions by observing and mimicking expert demonstrations (Hua et al., 2021). LfD has been employed in domains where defining performance metrics through traditional programming is challenging, such as robotic manipulation (El Zaatari et al., 2021; Li et al., 2023; Puranic et al., 2022; Sun et al., 2023; Wang et al., 2021), autonomous vehicle (Savari and Choe, 2021), and unmanned aerial vehicles (Bansal et al., 2016). Training reinforcement learning controllers to promptly grasp the connection between actions and responses is crucial for systems characterized by significant time delays. Given their reliance on gradient-based training methods, it can reduce the number of training iterations and ensure that the iteration direction of reinforcement learning network parameters remains accurate.

In this study, we develop a training strategy where LfD is applied in the training of deep reinforcement learning controllers for systems with large time delays. An expert-guided reinforcement learning controller training strategy is proposed. A PID controller is utilized as the expert to provide initial behavior demonstrations under a system with large time delays, to ensure the training efficiency. Then, a superior PID controller with a hypothetical trajectory is adopted to further enhance the performances of the reinforcement learning controller. The availability and the advantages of this two-stage training strategy is demonstrated by two cases, including one literature case and one industrial case.

2. Preliminaries

Reinforcement Learning (RL) is a branch of machine learning that enables an agent to learn optimal strategies through trial and error by interacting with an environment. As shown in Fig. 1, the agent receives the state of the environment s_t as input at each timestep t . Based on this state; the agent selects an action a_t . Subsequently, the environment responds with the next state s_{t+1} and the corresponding immediate reward r_t . A state is a description of the environment at a specific moment, while an action represents a behavior that the agent chooses based on a given state. The reward serves as immediate feedback for the action taken. The goal of reinforcement learning is to discover a policy through interactive processes that maximize the cumulative reward obtained by the agent during training. This framework aims to solve the Markov Decision

Process (MDP), characterized by a state space s , an action space a , a reward function $R(s, a)$, state transition probabilities $P(s'|s, a)$, and a discount factor γ . A policy π , defined as $\pi(a|s)$ maps given states s to actions a . The core objective in reinforcement learning is to find the optimal policy π^* that maximizes the total expected return:

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | \pi \right] \quad (1)$$

Among the various algorithms in reinforcement learning, the Deep Deterministic Policy Gradient (DDPG) is a prominent variant that integrates policy gradient methods with function approximation techniques, making it suitable for problems with continuous action spaces. DDPG employs deep learning models to approximate the policy function (Actor) and the value function (Critic), as shown in Fig. 1. DDPG employs an Actor neural network to determine a deterministic policy, which directly yields the optimal action for a given state. Simultaneously, a critic neural network evaluates the value function of the current policy, considering both the state and the action. These concepts can be expressed as follows: $\pi_{\theta}(s)$ represent the policy function, where θ are the parameters of the Actor neural network, and s represents the state; $Q_{\phi}(s, a)$ represents the action-value function, where ϕ are the parameters of the Critic neural network, and a represents the action.

The actor neural network yields a deterministic action $a = \mu_{\theta}(s)$, with a given state s , where θ is the neural network parameters. The objective of the actor is to maximize the estimated Q-value by the critic network, updating the policy gradient as follows:

$$\nabla_{\theta} \mathbf{J} = \mathbb{E}_{s \sim \rho^{\beta}} \left[\nabla_{\theta} \mu_{\theta} \nabla_{\theta} \mathbf{Q}_{\phi}(s, \mathbf{a})|_{\mathbf{a}=\mu_{\theta}(s)} \right] \quad (2)$$

where ρ^β denotes the state distribution according to the behavior policy β .

The critic neural network evaluates the Q-value $Q_{\phi}(s, a)$ for the current policy, given a state s and an action a , with ϕ as the network parameters. The training objective for the critic is to minimize the mean squared error between the predicted Q-values and the target Q-values:

$$L(\phi) = \mathbb{E}_{s,a,r,s' \sim D} [(\mathbf{Q}_\phi(s, a) - y)^2] \quad (3)$$

The target Q-value y is calculated using the target network parameters ϕ' and θ' to stabilize learning:

$$y = r + \gamma Q_{\phi'}(s', \mu_{\theta'}(s')) \quad (4)$$

The input to the Actor neural network is the current state s and the value function of the state-action pair $Q_\phi(s, a)$ provided by the critic network, while the output is a deterministic action a . The Critic network’s input includes the state s , the action output by the Actor, and the

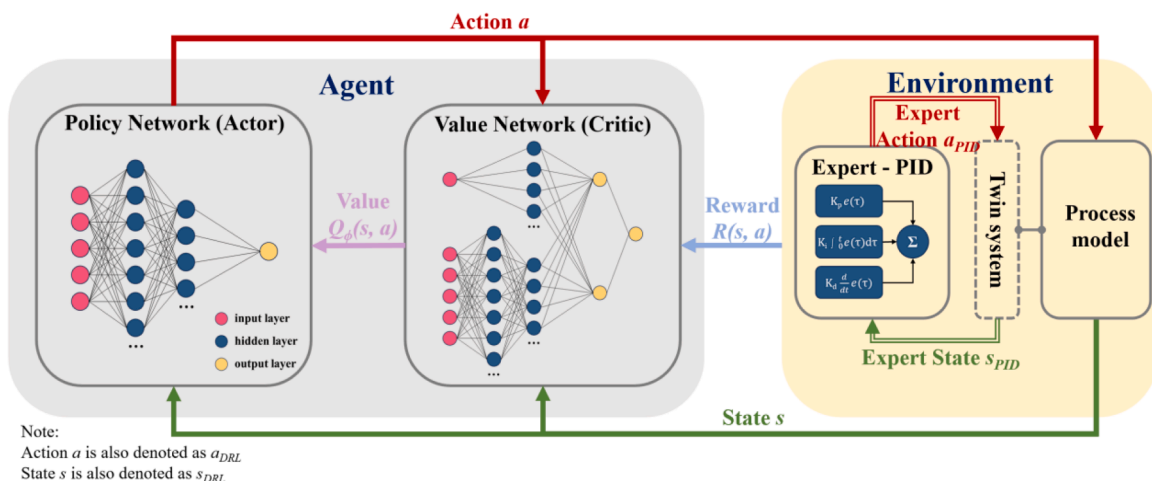


Fig. 1. Structure of the DDPG algorithm and the illustration of expert-guided reinforcement learning controller training strategy.

immediate reward $R(s, a)$ provided by the environment, and the output is the value function $Q_\theta(s, a)$ that estimates future rewards.

In reinforcement learning, the policy function $\pi(s)$ represents the mapping from a given state s to a selected action a , dictating the agent's decision-making strategy. The value function, on the other hand, evaluates the expected cumulative reward starting from a given state (or state-action pair) and following a particular policy thereafter. While the policy function governs the agent's behavior, the value function provides feedback on how desirable certain states or actions are, thus guiding the learning and optimization of the policy.

The Actor-Critic algorithm has been widely applied to a variety of complex decision problems and control tasks, including robotic control, gameplay, and resource management. In chemical reaction process control, minor variations in reaction conditions (such as temperature, pressure, and material feed rates) can significantly impact product quality and yield. Traditional control strategies may struggle with such highly nonlinear and multivariable systems. The Actor-Critic algorithm has the potential to be applied to such complex process control tasks.

3. Methodology

3.1. Two-stage expert-guided training strategy

This paper introduces a two-stage expert-guided strategy tailored to train deep reinforcement learning (DRL) controllers for processes with large time delays, as shown in Fig. 1. For systems with large time delays, the control variable cannot quickly reach the set point, when a new set point is given. If the reward is determined directly from the deviation between the control variable and the set point, the critic of the DRL controller cannot truly evaluate the action from the actor of the DRL controller. Thus, the proposed strategy introduces an expert, a PID controller in this study, into the training environment for delivering accurate rewards. The choice of a PID controller as an expert is due to its widespread application and proven stability in industrial control systems. During the training process, the expert (the PID controller) is operating parallelly with the DRL controller, as shown in Fig. 1. The PID controller is used in a twin system of the process model. In each episode, both systems are provided with the same initial state and set point, then the DRL controller and the PID controller will yield their action a_{DRL} and a_{PID} respectively. With these two actions, two states s_{DRL} and s_{PID} can be observed. Then, the rewards for the action a_{DRL} can be determined considering the state deviations between the DRL controller and the PID controller. By emulating the behavior of PID controllers in each episode,

the DRL controller can learn and adapt to the environment more quickly, overcoming the challenges posed by large time delays. Furthermore, to enhance the robustness of the training strategy, two training strategies are developed, named "learning the expert" and "surpassing the expert", as shown in Fig. 2. It is important to clarify that the transition from the first to the second training stage is performance-based. During the first stage training, agents are saved when their average rewards exceed a predefined threshold. The top three to five performing agents are then selected for the second stage training, based on the performance in few tests. Two stages are achieved by two different reward generation methods by the expert, which are introduced in the following.

3.1.1. Learning the expert

In the first stage, learning the expert, the reference trajectory generated by a PID controller is used to guide the training of the reinforcement learning controller. Specifically, in each training episode, a new set point for the control variable is randomly generated, then the PID controller will generate a reference trajectory of the control variable, as shown by the dashed curve in Fig. 2(a). A hypothetical control variable trajectory by the DRL controller is also shown in Fig. 2(a) by a solid curve. The deviation between two curves is defined as $error_{DRL-PID}$, which can be obtained by Eq. (5).

$$error_{DRL-PID} = \text{abs}(T_{DRL} - T_{PID}) \quad (5)$$

where T_{DRL} and T_{PID} are the current control variables via the DRL and PID controllers, respectively. The PID and DRL controllers operate synchronously, meaning that each point of the PID and DRL trajectory are obtained synchronously in each training episode. Then, the reward R can be obtained by a piecewise linear function, as shown in Fig. 3(a), where δ and γ ($\gamma > 1$) represent the threshold and the reward coefficient. When the $error$ is larger than δ , R will be equal to $\delta - error$. Otherwise, R will be $\gamma(\delta - error)$. R increases monotonously with the decreasing $error$, while the increasing ratio will be larger if the $error$ is smaller than the threshold δ . The coefficient γ is 10 in all studied cases, which emphasizes precise tracking when the agent is close to the expert trajectory. The threshold δ is 0.1 in all studied cases. The pseudocode of the reward calculation in this stage is given in Table 2. The mathematical equation can be obtained by Eq. (6).

$$R = \begin{cases} \gamma * (\delta - error) + P, & \text{if } error \leq \delta \\ \delta - error + P, & \text{if } error \geq \delta \end{cases} \quad (6)$$

where P is a penalty term, which is introduced in the following.

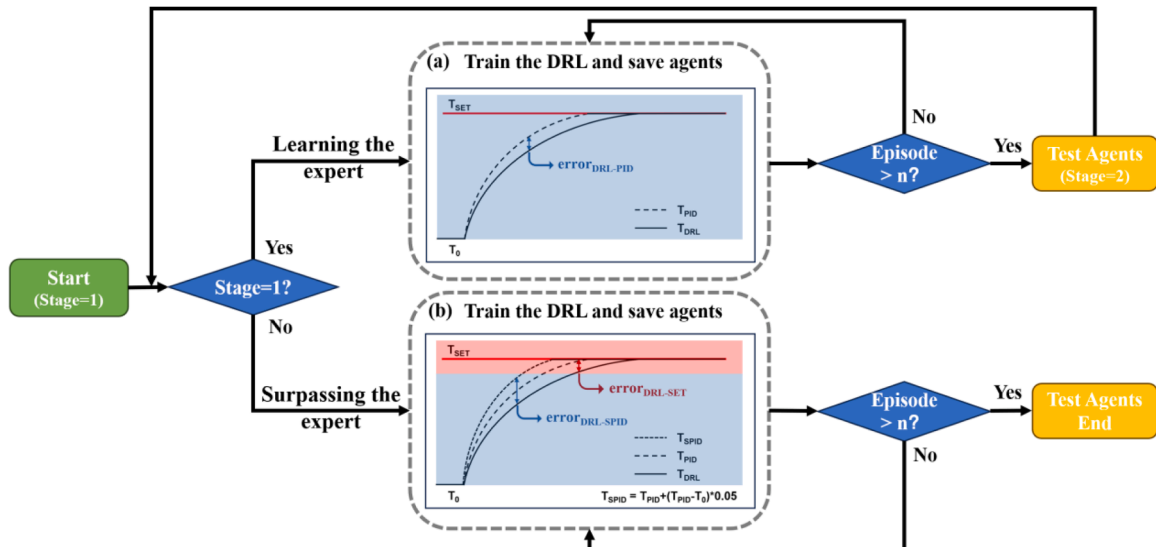


Fig. 2. A schematic of the proposed training strategy.

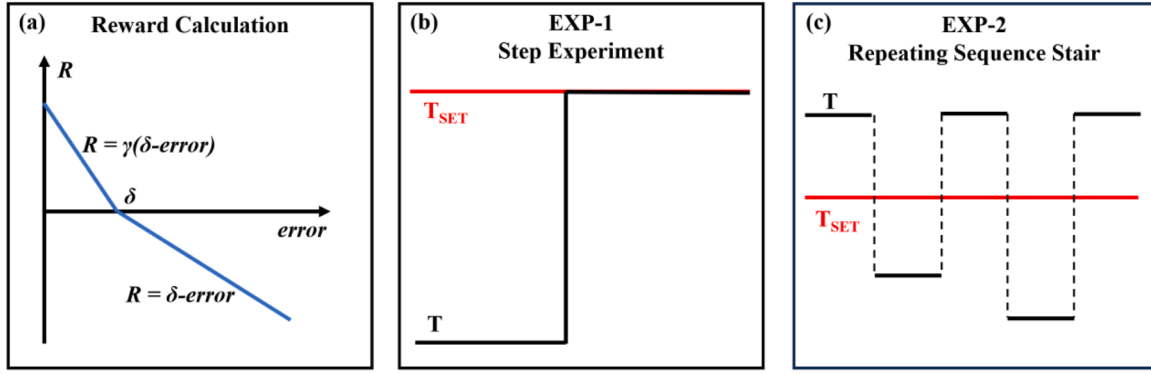


Fig. 3. Schematic diagram of saving agent rules and experiments.

Table 2

Reward algorithm pseudocode.

Reward setup pseudocode for each time step
Input T_{SET} , T_{PID} , T_{DRL} $error = error_{DRL-PID}$ OR $error_{DRL-SPID}$ OR $error_{DRL-SET}$ if $error \geq \delta$ $R = \delta - error$ else $R = \gamma * (\delta - error)$ end

Throughout the training process, a series of agents (DRL controllers) are automatically saved, when the average reward exceeds a given reward threshold η . The agents are named as FS-Agent-X, where X represents the X_{th} episode. Before proceeding to the second stage, all saved agents are evaluated through two experiments, a step experiment (EXP-1) and a repeating sequence stair experiment (EXP-2), as shown in Figs. 3(b) and 3(c). The IAE for the DRL and PID controllers are recorded for both EXP-1 and EXP-2. According to the IAE results, the top three to five potential DRL controllers are found. The corresponding pseudocode is shown in Table 3. This selection process ensures that only the most suitable agents proceed to the next training stage, thus enhancing the overall training efficiency.

3.1.2. Surpassing the expert

In the second stage, surpassing the expert, a hypothetical trajectory based on the PID trajectory is generated, namely superior PID or SPID trajectory. Similar to the first stage, the PID controller will generate a reference trajectory of the control variable for any given set points. Then, a SPID trajectory can be obtained by Eq. (7). The SPID trajectory is considered as a trajectory that responds faster than the PID, as shown by the dashed curve in Fig. 2(b). The deviation between DRL and SPID is defined as $error_{DRL-SPID}$, which can be obtained by Eq. (8). The SPID trajectory exhibits a higher tendency to overshoot compared to the PID trajectory, as it is derived from the PID trajectory and is designed to achieve a faster response. Thus, when the control variable is close to the

set point (i.e. the red part in Fig. 2(b)), the deviation will be directly obtained by T_{DRL} and T_{SET} . Specifically, when the deviation between T_{DRL} and T_{SET} is within 5 % of T_0 and T_{SET} , the DRL controller will directly adjust its control strategy according to set points rather than the expert of SPID, which can be obtained by Eq. (9). The reward R is obtained by the deviation $error$ and Eq. (6). Similar to the first stage of training, a series of agents, namely SS-Agent-X, are automatically saved, when the average reward exceeds a given reward threshold η . The procedure of Table 3 is carried out to select three DRL controllers from all saved agents.

$$SPID = T_{PID} + (T_{PID} - T_0) * 0.05 \quad (7)$$

Second stage-1:

$$error_{DRL-SPID} = abs(T_{DRL} - T_{SPID}) \quad (8)$$

Second stage-2:

$$error_{DRL-SET} = abs(T_{DRL} - T_{SET}) \quad (9)$$

By introducing Eq. (8) and Eq. (9), the training stage is divided into two phases. The first phase follows the SPID curve to achieve better control performances than the PID controller. This phase is essential for large-delay systems, without it, the long dead time would yield sparse and highly delayed feedback, forcing the actor to take large exploratory steps that can easily excite oscillatory modes. By constraining exploration to a feasible reference, the first phase establishes a well-conditioned starting policy, after which the second phase can proceed stably and efficiently. In the second phase, the focus shifts to directly tracking the set value to obtain stable control performance, reducing the likelihood of overshooting and oscillations. In addition, it is worth noting that the training method does not necessarily need to start from the first stage. For a relatively simple problem, skipping the first stage and directly entering the second stage is feasible.

3.2. Training termination

To ensure the stability of the reinforcement learning system throughout the learning process, we also introduce a protective mechanism capable of terminating the training, as depicted in Fig. 4. The agent includes an interface named "Termination", implemented after two modules determining whether the termination conditions are met. Suppose the control curve surpasses either the upper or lower limit ($T_{SET} + upper\ limit$ or $T_{SET} - lower\ limit$). In that case, the episode will be immediately terminated, and a substantial penalty P will be incorporated into the reward, which can be obtained by Eq. (10). The termination mechanism enables the DRL to progressively learn the following principle: the control curve must not significantly deviate from the setpoint, necessitating its control within a reasonable range. This mechanism is essential for the application of reinforcement learning in industrial production. The threshold should reflect practical industrial

Table 3

Pseudocode of agent selection rules.

Pseudocode of agent selection rules
Load all M agents, $Agent_Result = []$ Simulate the PID controller in EXP-1 and EXP-2 to yield IAE_{PID}^1 , IAE_{PID}^2 for agent m in M: Simulate the agent m in EXP-1 and EXP-2 to yield IAE_{DRL}^1 , IAE_{DRL}^2 Calculate $\frac{IAE_{PID}^1 - IAE_{DRL}^1}{IAE_{PID}^1} + \frac{IAE_{PID}^2 - IAE_{DRL}^2}{IAE_{PID}^2}$, and save it in $Agent_Result$ end Sort $Agent_Result$ in descending order Select Top 3–5 from $Agent_Result$

tolerance limits, ensuring the controller's learning process remains within operational constraints. The specific effects of this setup are demonstrated in Fig. 4, which displays the case of exceeding the upper or lower limit. The episode will be instantly halted once the control variable reaches the position marked by a red cross.

$$P = \begin{cases} -1000, & \text{if } T_{DRL} \geq T_{SET} + \text{upper limit or } T_{DRL} \leq T_{SET} - \text{lower limit} \\ 0, & \text{if } T_{SET} - \text{upper limit} < T_{DRL} < T_{SET} + \text{lower limit} \end{cases} \quad (10)$$

4. Case study

A benchmark case and an industrial case are presented in this section to demonstrate the availability and efficiency of the proposed two-stage export-guided training strategy. The Integral Absolute Error (IAE) is adopted to quantify the performance of controllers. IAE represents the cumulative value of the absolute deviation $e(t)$ between the setpoint and the actual value of the control variable over time, which is defined as follows:

$$IAE = \int_0^{\infty} |e(t)| dt \quad (11)$$

4.1. Benchmark

The control of nonlinear valves with friction and time delays from Siraskar (2021) is studied as a benchmark case. Siraskar utilized a segmented training approach, dividing a delay of 2.5 s into seven stages and incrementally increasing the delay. There were also incremental increases in other parameters, such as friction. This approach is metaphorically referred to Graded Learning, inspired by observing how human instructors structure the coaching of a new skill for apprentices: foundational level skills are taught first, and once a certain level of proficiency is attained, the apprentice progresses to the next level to tackle slightly more complex problems.

A deep reinforcement learning control system is designed and trained by the methodology outlined in Section 3. The control system structure and the corresponding hyperparameters are presented in the Supporting Information. It should be noted that the control object of this DRL controller is the water tank, and its operating variable is the valve opening. *State* is defined as follows:

$$\text{State} = \begin{bmatrix} \text{Controlled Signal} \\ \int_0^t e_k dt \end{bmatrix} \quad (12)$$

During the 1000 episodes of the first stage of training, the learning rates for both the actor and the critic are set at $5e-4$. FS-Agent-1000 is selected for the second stage of training. In the 1000 episodes of the second stage of training, we reduce the learning rates to $3e-4$. Then, SS-Agent-307 is selected as the best DRL controller to be evaluated by two experiments from the literature (Siraskar, 2021). In Test-1, all controllers' initial points and set points are set to 0 and 100, respectively. In Test-2, the initial point of all controllers is also set to 0, while the set point is 100, 25, 75, and 25 within 0–750 s, 750–1500 s, 1500–1750 s, and 1750–2000 s. The simulation time of both two experiments are 2000 s. The results of the PID controller, the obtained DRL controller, the DRL controller (Siraskar, 2021) are shown in Fig. 5, and the IAE values are shown in Table 4. Since we do not have access to the original data of the literature (Siraskar, 2021), the green curve displayed is derived from digitizing the image in the literature using the software Origin. Both the DRL controller via this paper and Siraskar (2021) outperform the PID controller in terms of adjustment speed and overshoot, as shown in Fig. 5 (a) and Fig. 5(b). It can also be seen that the adjustment speed of the DRL controller via this paper (blue curve in Fig. 5) is faster than the DRL controller of Siraskar (2021) (green curve in Fig. 5). Fig. 5(a) also shows a considerable gap between the green curve and the set value for Test-1. It indicates that the DRL controller of Siraskar (2021) might sacrifice the tracking performance to avoid overshooting. As for the DRL controller of this paper, it might slightly overshoot, but it can reach the set value and maintain stability near the set value. Fig. 5(b) shows two DRL controllers exhibiting some oscillation during the latter two step-down experiments, but the oscillation's extent is insignificant. According to the IAE values shown in Table 4, the IAE values for Test-1 and Test-2 via the DRL controller of this paper are 61.9 % and 25.0 % smaller than the ones via the DRL controller of Siraskar (2021), and 56.7 % and 56.4 % smaller than the ones via the PID controller. Although the time delay of this benchmark case is not as significant as those in process industries, the DRL controller derived from the two-stage training method shows better performances than the one by multi-stage training via incrementally increasing time delay.

4.2. Industrial case: fluidized bed reactor temperature control

The fluidized bed polyethylene process is one of the most common

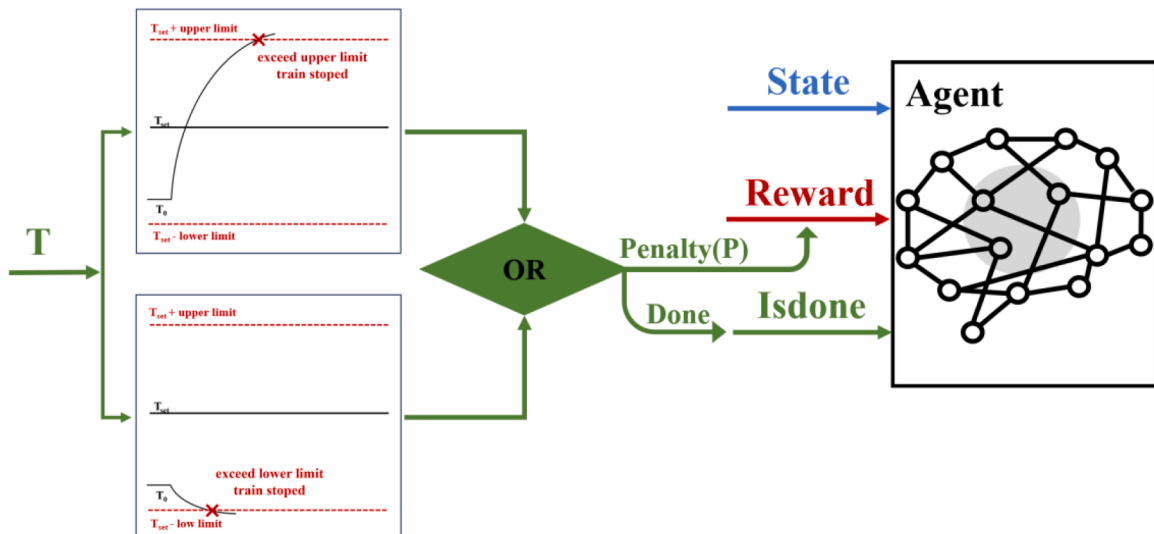


Fig. 4. Training termination mechanism.

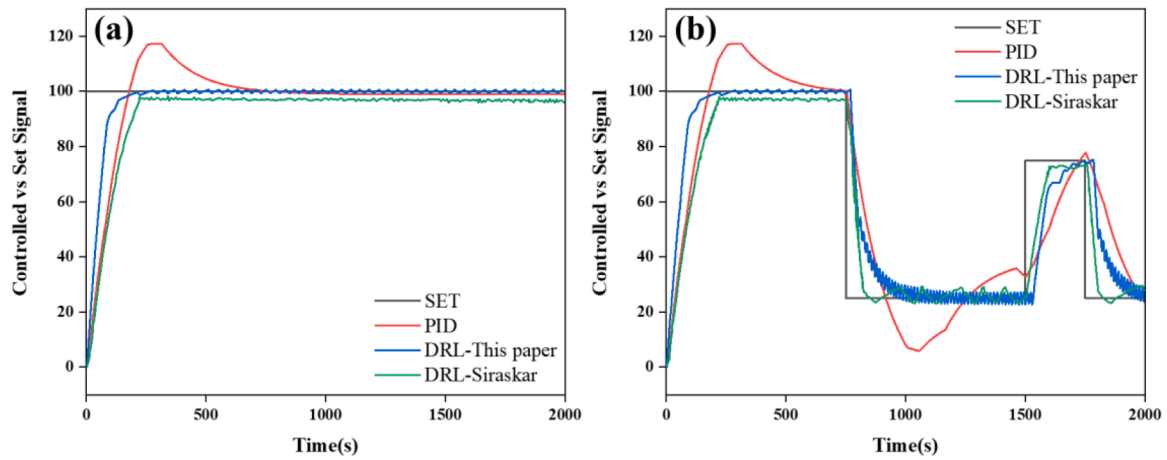


Fig. 5. Comparison of the two-stage expert-guided and multi-stage training methods(Siraskar, 2021). Note that the data of DRL-Siraskar is obtained via a digitalization tool of the software Origin from the literature figure.

Table 4

IAE values of three controllers for benchmark case.

Item	IAE-Test-1	IAE-Test-2
PID	13,168	33,580
DRL-Siraskar	14,970	19,510
DRL-This paper	5700	14,630

polyethylene processes. The ethylene polymerization reaction is significantly exothermic, necessitating appropriate thermal management to control the reaction temperature. Temperature control is crucial as temperature directly affects the properties and productivity of the polymer. The heat exchange system of the fluidized bed polyethylene process is shown in Fig. 6, which employs a fouling prevention heat exchange scheme. The recycled gas is cooled by the demineralized water (DW) but not the cooling water (CW). As shown in Fig. 6, the cooling system for the recycled gas involves a shell and tube heat exchanger

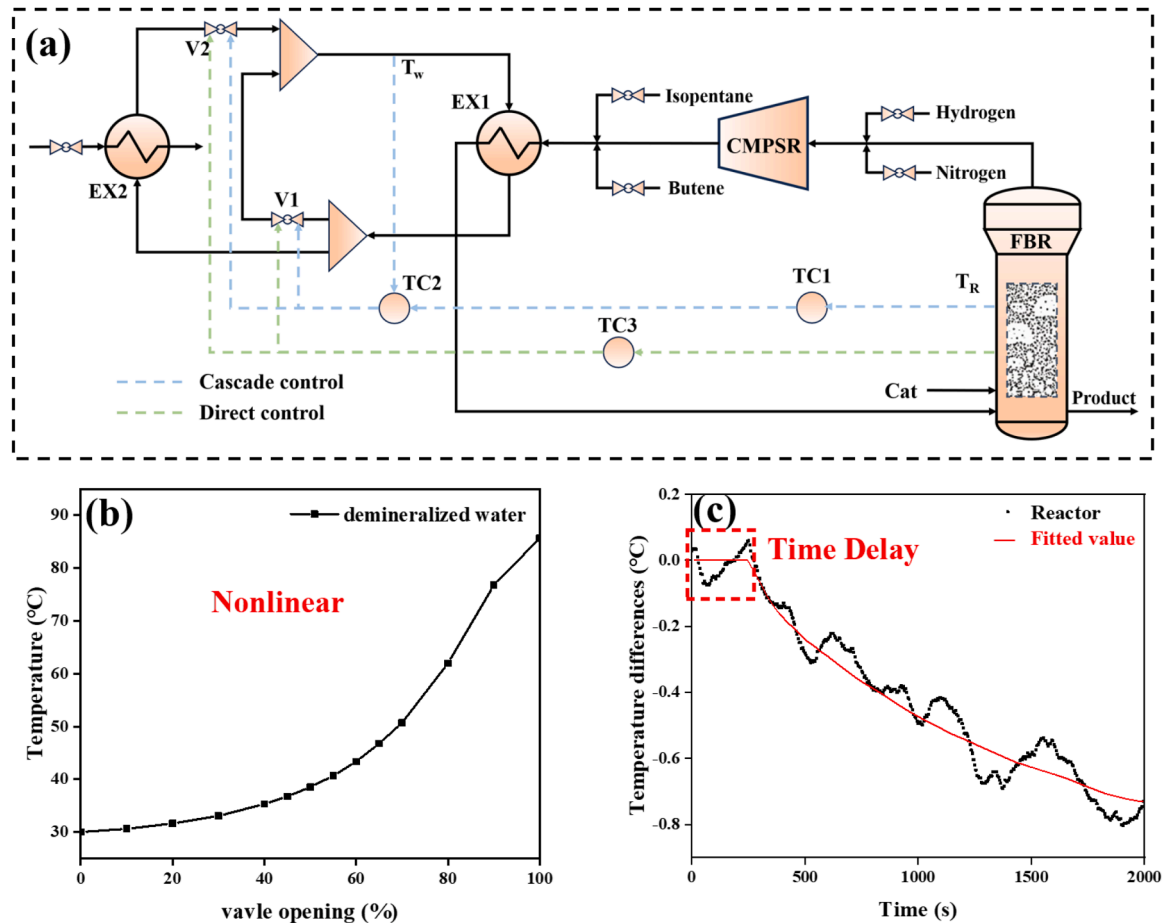


Fig. 6. Schematic diagram of (a) fluidized bed polyethylene process and its temperature control schemes. (b) nonlinear characteristic of the control valve, and (c) time delay.

(EX1) and a plate heat exchanger (EX2). The demineralized water receives heat from the recycled gas via EX1 and discharges heat to the cooling water via EX2. The process can be managed using a direct control scheme (TC3 of Fig. 6), where the reactor temperature is controlled by adjusting the proportion of demineralized water bypassing to heat exchanger EX2. It is achieved by the opening adjustment of valves V1 and V2, the sum of whose openings always equals 100 %. V1 controls the flow of bypass demineralized water, while V2 controls the flow of demineralized water exchanging heat with the cooling water. A wider opening of V1 corresponds to a higher temperature of the demineralized water. In the direct control scheme, the controller receives the temperature signal from the reactor, subsequently adjusting the valve openings to modulate the temperature of the demineralized water and achieve the desired reactor temperature control. In the cascaded control scheme (TC1 and TC2 of Fig. 6), the main loop variable is the reactor temperature, and the secondary loop variable is the demineralized water temperature.

The difficulty of precise temperature control is due to two system characteristics. Firstly, when the valve opening is below 60 %, the temperature of the demineralized water shows minimal fluctuation with changes in the valve opening, as shown in Fig. 6(b). When the valve opening exceeds 60 %, the temperature changes significantly with an increase in the valve opening, indicating that the heat exchange system exhibits non-linearity. Furthermore, due to the lag associated with valve adjustment, fluid transmission, and heat exchange, a notable delay can be observed during the control process, as shown in Fig. 6(c). Temperature changes in the demineralized water require substantial time to impact the reactor temperature. Such a delay in temperature control can potentially lead to fluctuations and even oscillations in the reactor temperature, presenting challenges in designing effective reactor temperature control schemes.

4.2.1. Control system

A traditional PID-PID cascade control scheme for the temperature control of Fig. 6(a) is shown in Fig. 7(a), where $G_p(s)$ and $G_s(s)$ are the process model of the main loop and the secondary loop, respectively. Both $G_p(s)$ and $G_s(s)$ are from the closed-loop identification experiment. The model is as follows:

$$G_p(s) = \frac{0.2992}{1267s + 1} e^{-264.8s} \quad (13)$$

$$G_s(s) = \frac{0.66404}{48.002s + 1} \quad (14)$$

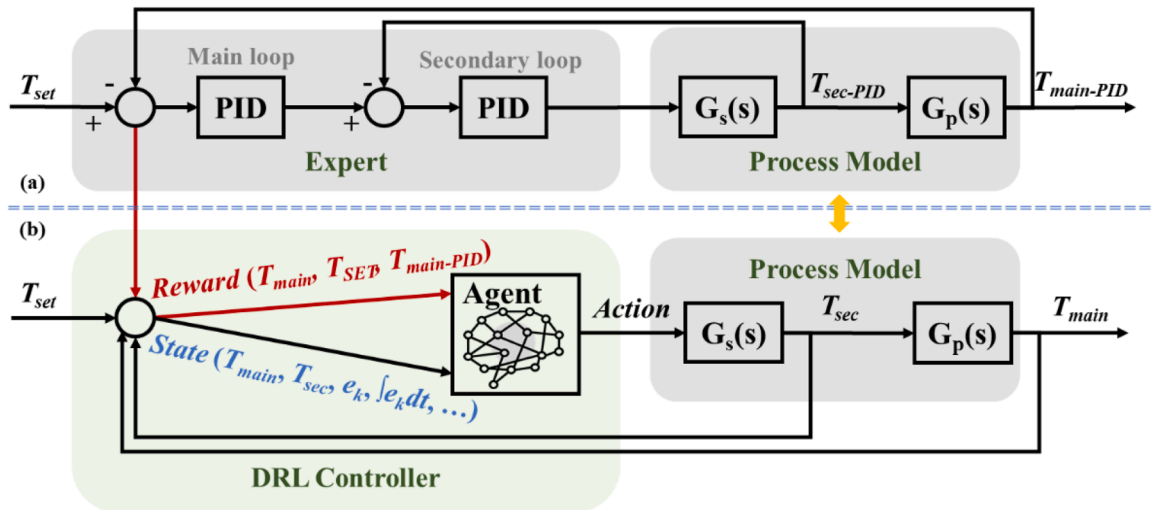
It is worth noting that the secondary loop process models have strong nonlinearity, which cannot be represented by the transfer function of Eq. (14). An additional nonlinear term is introduced to capture the dynamic behavior. Specific statements can be found in Supporting Information 2.5

The main loop and secondary loop process models have been introduced in our previous work (Hong et al., 2024). The parameters for PID-PID cascade control are obtained through Simulink's PID tuning module, shown in Table 5. The PID-PID control scheme cannot efficiently deal with the nonlinearity introduced above. When dealing with systems that exhibit complex dynamics, high order, time delays, and nonlinear characteristics, traditional PID control may encounter several challenges (Abbasi et al., 2019; Gaing, 2004; George and Ganesan, 2022), such as difficulties in parameter tuning, increased oscillations, and poor robustness. In our previous work, we developed a DRL controller for the secondary loop (Hong et al., 2024). We employed a PID-DRL cascade control scheme, which performed better than the PID-PID cascade control scheme. However, the training strategy of the secondary loop was unsuitable for the main-loop control with a large time delay. To further improve the control performance, we aim to design a standalone DRL controller, which can deal with the large time delay of the gas-phase fluidized bed temperature control. Furthermore, we also design a main loop DRL controller for the DRL-DRL cascade control scheme to address potential disturbances in the secondary loop. Fig. 7(b) illustrates the control scheme of the standalone DRL controller. The DRL controller receives the set point T_{set} for the main loop and the current T_{main} and T_{sec} . The DRL controller determines an action for the control valve based on the state, whose setup is introduced in the following section. Then, the process model of $G_p(s)$ and $G_s(s)$ will update T_{main} and T_{sec} of the control system. The training strategy is also shown in Fig. 7. The PID-PID cascade controller is used as an expert during the DRL training, as the training strategy introduced in Section 3. Note that

Table 5

PID parameters of the main and secondary loops.

Parameter	P	I	D	N
Main loop	5.4944	0.0059	274.8486	0.0030
Secondary loop	0.6234	0.0662	-8.4931	0.0734



Note: Red lines are only used during training.

Fig. 7. Structure of the PID-PID cascade control scheme and the Deep reinforcement learning controller.

the DRL-DRL cascade controller scheme is similar to the PID-PID cascade control scheme, thus it is not introduced in this paper.

In this study, the termination threshold is set at ± 2.5 °C from the setpoint. This range is based on practical industrial standards, where temperature deviations within ± 2.5 °C in fluidized bed reactor operations are generally regarded as acceptable for maintaining product quality and process safety. Embedding this constraint in the reward structure ensures that the agent learns to operate within a realistic and safe control envelope.

4.2.2. DRL control setup

States, actions, and rewards are critical for the Markov decision process and reinforcement learning, significantly impacting training efficiency and controller performances. For the DRL controller (i.e., the agent) of Fig. 7(b), we design two states *State_A* and *State_B*, which include four observations and five observations, respectively. *State_A* includes four observations: the demineralized water temperature T_{sec} , the reactor temperature T_{main} , the temperature error e_k , and the cumulative error $\int e_k dt$. The demineralized water temperature T_{sec} directly affects the reactor's temperature. It is the main means of control. The reactor temperature T_{main} is the control variable that the system needs to control precisely. The temperature error e_k is the derivation between the reactor temperature T_{main} and the set point T_{set} . The cumulative error $\int e_k dt$ is the total sum of e_k . These four observations are critical indicators for assessing control performance and adjusting control strategies, providing feedback on immediate control effects and long-term stability. *State_A* is defined as follows:

$$State_A = \begin{bmatrix} T_{sec} \\ T_{main} \\ e_k \\ \int e_k dt \end{bmatrix} \quad (15)$$

where e_k equals to $T_{set} - T_{main}$.

To enhance the DRL controller performance, we introduce another observation T_{stable} , which represents the steady-state reactor temperature corresponding to the demineralized water temperature. For example, when the demineralized water temperature is 40 °C, the reactor temperature will reach 85.43 °C if the demineralized water temperature is kept constant. By incorporating T_{stable} into *State_B*, we aim to have the deep reinforcement learning controller predict the steady-state reactor temperature based on a given demineralized water temperature. *State_B* is defined as Eq. (16). A formulation between T_{stable} and T_{sec} is derived from the simulation results via Simulink.

$$State_B = \begin{bmatrix} T_{sec} \\ T_{main} \\ e_k \\ \int e_k dt \\ T_{stable} \end{bmatrix} \quad (16)$$

$$T_{stable} = 0.00000071T_{sec}^4 - 0.00016863T_{sec}^3 + 0.01432000T_{sec}^2 - 0.21400000T_{sec} + 80.05099000 \quad (17)$$

Similar to industrial fluidized bed reactor temperature control systems, the action from the deep reinforcement learning (DRL) controller is the valve opening. At each sampling time k , the DRL controller emits a control signal A_k , which is predicted by the actor-network. In the training process's code, the valve's initial opening is set at 50 %, which means the valve opening range is $[-50, 50]$. It is a common state of valve opening during the production, corresponding to a demineralized water temperature of 38 °C and a reactor temperature of 85 °C. In addition, the reward is obtained according to the set point T_{set} , the control variable of the DRL controller T_{main} , and the controller variable of the expert $T_{main-PID}$, which is introduced in Section 3.1. In each complete training process, referred to as an episode, the set value T_{set}

and the initial value T_{main} and $T_{main-PID}$ of the reactor temperature are randomly generated. It allows the deep reinforcement learning controller to learn various control trajectories that are applicable to different operating conditions. The ranges for T_{set} , T_{main} and $T_{main-PID}$ are all set 83~87 °C.

In summary, the DRL controller establishes a direct mapping from the observation space to the control action. The control policy can be mathematically represented as:

$$A_k = f(S_k) \quad (18)$$

where S_k denotes the current observation state (either *State_A* or *State_B* as defined), and A_k denotes the corresponding control action, i.e., the valve opening adjustment at time step k . This mapping is realized through a deep neural network (the Actor network) trained under the expert-guided two-stage reinforcement learning framework.

4.2.3. Result and analysis

According to the two-stage expert-guided training strategy introduced in Section 3 and the DRL controller setup, we design two standalone DRL controller and a DRL-DRL cascade controller (namely C-DRL-DRL). Two standalone DRL controllers are S-DRL-A and S-DRL-B, which use *State_A* and *State_B*, respectively. The controllers obtained by the first and second stage training are n-DRL-A/B-FS and S-DRL-A/B-SS, respectively, as shown in Table 6. The C-DRL-DRL controller uses *State_B* for the main loop DRL controller, while using a state with three observations for the secondary loop DRL controller. The state of the secondary loop has been introduced in our previous work (Hong et al., 2024). Two controllers, the PID-PID cascade controller and the PID-DRL cascade controller from our previous work (Hong et al., 2024), are adopted for the comparisons in the following. The controller structure and the observation dimension of the DRL are all shown in Table 6. Table S5 of the Supporting Information shows the hyperparameters for all DRL controllers.

While our framework focuses on PID-PID cascade control as the expert, which aligns with mainstream industrial practice, we recognize the potential advantages of model-based Model Predictive Control (MPC) as a benchmark. To address this, we provide a supplementary analysis comparing DRL with MPC under EXP-1, EXP-2, and EXP-3 scenarios. Detailed results are presented in the Supporting Information (Section 2.6, Figure S8) for reference.

4.2.3.1. DRL controller comparisons. We design five experiments to validate the effectiveness of the DRL controllers introduced above. Experiment 1 (EXP-1) is a step response experiment where the reactor temperature is stepped from 85 °C to 86 °C, with a total simulation time of 10,000 s. We aim to study the controller performances regarding overshoot and response time, by EXP-1. Experiment 2 (EXP-2) is a reactor repeating sequence stair experiment, where a random fluctuation from $[+0.2, -0.5, +0.5, -0.6, +0.6]$ °C occurs every 2000 s and the set point keeps constant at 85 °C. EXP-2 evaluates whether the controller can maintain the temperature at the set point within 2000 s. In experiment 3 (EXP-3), a random fluctuation from the range of $[-10, 10]$ °C is added to the demineralized water temperature (the control variable of the secondary loop) every 60 s. To simulate real scenarios, experiment 4

Table 6
Controllers investigated in this study.

No	Controllers	Control structure	Observation dimension of DRL
1	S-DRL-A-FS	DRL	4
2	S-DRL-A-SS	DRL	4
3	S-DRL-B-FS	DRL	5
4	S-DRL-B-SS	DRL	5
5	C-DRL-DRL	DRL-DRL	5 + 3
R-1	C-PID-PID	PID-PID	—
R-2	C-PID-DRL	PID-DRL	3

(EXP-4) adds random disturbances of $[-0.05, 0.05\text{ }^{\circ}\text{C}]$ to EXP-1 every 100 s. In experiment 5 (EXP-5), similar random disturbances are added to EXP-2.

The experiment results via the seven controllers in Table 6 are shown in Fig. 8, while the IAE values are listed in Table 7. Additionally, ISE and ITAE can be found in the Supporting Information 2.7. Comparing two standalone DRL controllers (S-DRL-A-FS and S-DRL-B-FS) obtained via the first stage training and the C-PID-PID controller, it is found that the performance of the DRL controllers is close to the one of the C-PID-PID controller in EXP-1 and EXP-2. Although the first stage is the so-called learning the expert stage, the performance can be better than the expert (C-PID-PID), as shown in EXP-1. Besides, S-DRL-A-SS and S-DRL-B-SS, the controllers obtained after the second stage training, perform better than S-DRL-A-FS, S-DRL-B-FS, and C-PID-PID in EXP-1 and EXP-2. For example, in EXP-2, the IAE results via S-DRL-A-SS and S-DRL-B-SS are 23.4 % and 17.4 % lower than the ones S-DRL-A-FS and S-DRL-B-FS. It demonstrates the effectiveness of the so-called surpassing the expert stage. The DRL controller has achieved better control performance than the one from the first stage and the expert. Furthermore, comparisons between DRL controllers with four observations and five observations reveal that the addition of T_{stable} further increases the control efficacy of the DRL controllers. In the second training stage, the performance of S-DRL-B-SS exceeds that of S-DRL-A-SS by 15.5 % and 9.5 % in EXP-1 and EXP-2, respectively. It is because T_{stable} reflects the steady-state reactor temperature related to the demineralized water temperature T_{sec} . Inclusion of T_{stable} in the state enriches the understanding of the reactor dynamics for the DRL controller and enhances the learning capabilities of the DRL controllers.

However, all four standalone DRL controllers cannot deal with the fluctuation from the secondary loop in EXP-3. It is the inherent issue of the direct control structure. Conversely, a cascade control scheme exhibits better management of such disturbances. It is one of the reasons why cascade control schemes are predominantly employed in the studied gas-phase fluidized bed reactor. The C-DRL-DRL controller obtained in this study can overcome this issue. The IAE values for the C-DRL-DRL controller are smaller than those from our previous work by the C-PID-PID and the C-DRL-PID controllers. Notably, the IAE of C-DRL-DRL are 9.2 %, 4.2 % and 2.9 % smaller than that of C-PID-DRL controller, which only use one DRL controller. Moreover, the results of EXP-4 and EXP-5 indicate that the DRL controllers can maintain stable performance with disturbances, showcasing remarkable resistance to interference. The results of EXP-4 and EXP-5 can be found in the Supporting Information.

Considering the standalone DRL controller, the performance of S-DRL-B-SS is slightly better than that of C-DRL-DRL in two main loop experiments (EXP-1 and EXP-2). When the deviation is from the main loop, the direct control scheme is more straightforward than the cascade control scheme. The direct control scheme can perform better than the cascade control scheme, as demonstrated by the comparisons between the standalone PID controller and the PID-PID cascade control scheme.

Table 7

IAE values under experiments 1, 2, 3.

Item	EXP-1	EXP-2	EXP-3
S-DRL-A-FS	907.76	2192.09	1746.91
S-DRL-A-SS	750.04	1678.09	1546.89
S-DRL-B-FS	643.93	1839.23	1609.04
S-DRL-B-SS	633.02	1519.51	1463.69
C-DRL-DRL	655.75	1594.20	300.26
C-PID-PID	799.52	1824.45	513.59
C-PID-DRL	722.56	1664.19	308.54

The corresponding data can be found in Table S6 of the Supporting Information. However, it does not undermine the overall control performance of the C-DRL-DRL and its applicability to fluidized bed reactor control systems, as demonstrated in EXP-3 (disturbances from the secondary loop), where its performance significantly surpassed that of S-DRL-B-SS by 388 %. Thus, the slight advantage of S-DRL-B-SS on EXP-1 and EXP-2 becomes less significant, making C-DRL-DRL a more suitable controller.

4.2.3.2. Training strategy comparisons. We also adopt the multi-stage training and divides the time delay into segments, such as two, four, eight, and sixteen pieces. For instance, a time delay of 264.8 s divided into eight pieces results in training the DRL controller with an initial 33.1 s, progressively increasing by 33.1 s until the full delay is included. Figs. 9(a) and 9(b) illustrate the IAE results from each training stage for the step response experiment (EXP-1) and the repeating sequence stair experiment (EXP-2), respectively. Both experiments consistently increase IAE values with longer time delays, indicating higher system complexity and training difficulty. IAE values in EXP-1 and EXP-2 exceed 1000 and 1500, respectively, when the full delay is considered for two, four, and eight stages. They surpass the S-DRL-B-SS results represented by a black dashed line. In the sixteen-stage scenario, IAE values rise above 2000 at the 8th stage, leading to training termination.

The main difference between the proposed method and segmented training is that the proposed method considers the full-time delay and uses expert guidance to reduce training difficulties. This improves DRL controller performance and ultimately enables DRL controller performance to surpass the experts. Conversely, segmented training witnesses deteriorating performance as the stages progress. Segmented training relies on models refined from previous stages. Issues encountered during intermediate stages significantly impact subsequent training, as errors are propagated through segmented training. Dividing the time delay into more stages does not improve outcomes, as the results show. The training process faces similar difficulties if the time delays are identical. This is evidenced by the IAE results, which are comparable across different methods when the time delays are the same. With its two-stage expert-guided approach, the proposed method effectively addresses time delay issues at lower computational costs. Therefore, segmented training may be impractical for systems with significant time delays due

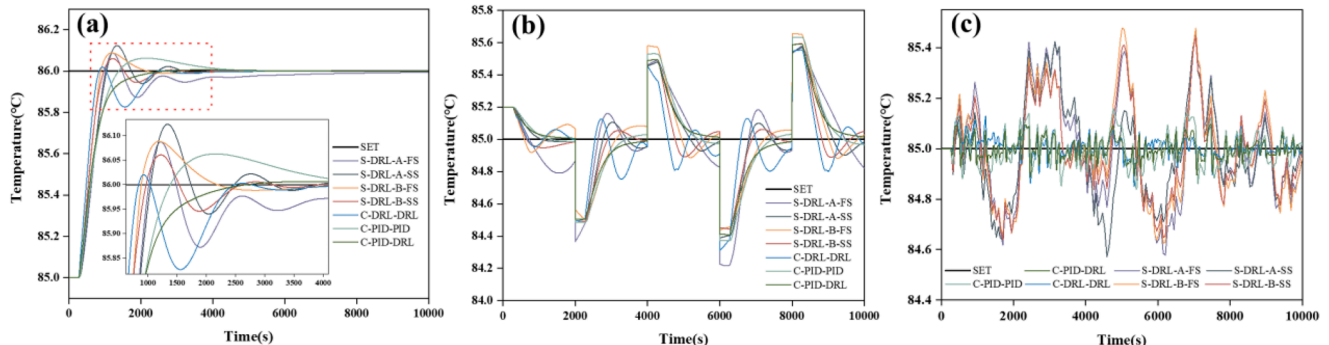


Fig. 8. Curves of experimental results. (a)EXP-1; (b) EXP-2; (c) EXP-3.

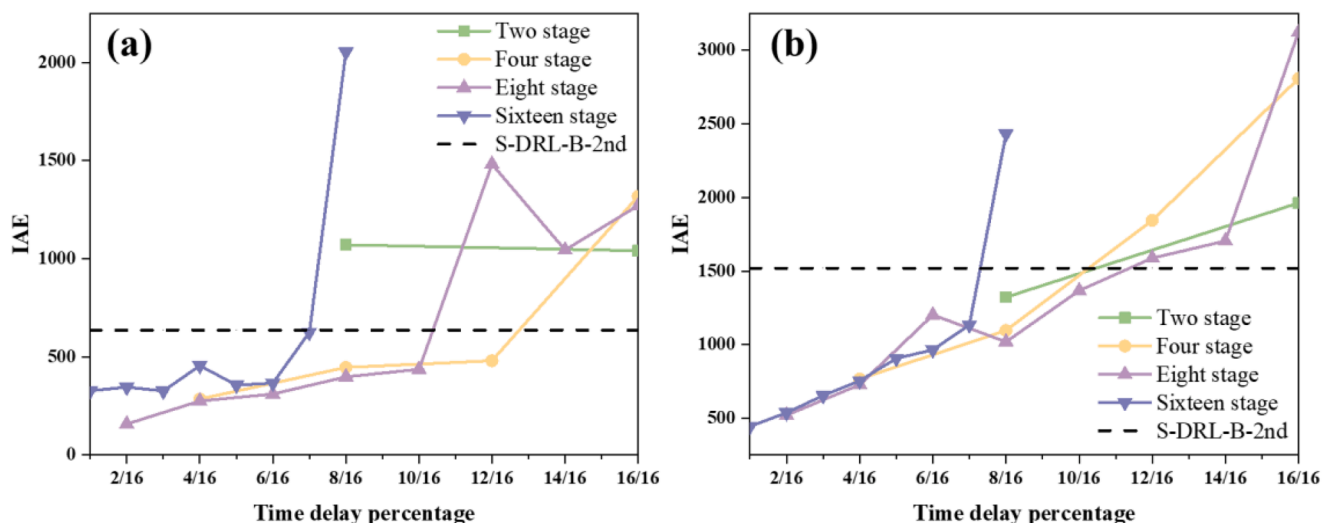


Fig. 9. IAE curves of segmented training. (a)EXP-1; (b)EXP-2.

to the substantial instability and excessive computational costs.

To substantiate the claimed computational advantage, we recorded wall-clock training times on identical hardware (AMD Ryzen 7 5800H, 32 GB RAM) using MATLAB R2022b Reinforcement Learning Toolbox. The proposed expert-guided two-stage curriculum required roughly two hours per 1 000 training steps in each stage and one additional hour for automated agent evaluation, so a full training run converged in about six hours while delivering lower tracking error and greater stability than a conventional two-segment baseline completed within the same time budget. Further partitioning the 264.8 s delay into four or eight segments stretched total training times to approximately eleven and twenty-one hours, respectively, and control performance deteriorated accordingly. When the delay was split into sixteen segments, cumulative runtime exceeded 24 h and the run failed to converge because error accumulation destabilized later stages. These observations confirm that the advantages of leveraging expert trajectories to guide exploration. The proposed method avoids repeated warm-starts and error propagation, thereby achieving superior closed-loop performance with equal or markedly lower computational cost than the segmented-training method.

5. Conclusions

We develop a two-stage expert-guided reinforcement learning framework for training controllers in large time-delay systems. An expert is introduced into the RL environment to provide references for rewarding the agent's actions. The training consists of two stages: learning from and surpassing the expert. A PID controller is adopted as the expert in the first stage, while a hypothetical controller is used in the second stage. The trajectories generated by the expert provide a practical and informed baseline, which significantly expedites the training process. The two-stage method facilitates DRL controllers' initial acquisition of PID control strategies in the first stage. It empowers them to surpass conventional PID controllers in the second stage.

Two case studies, including an industrial case with a substantial time delay of 264.8 s, demonstrate the effectiveness of the proposed two-stage expert-guided training strategy. The DRL controllers trained by the proposed method perform better than those trained through a literature segmented method. The results also demonstrate that after surpassing the expert stage, the obtained DRL controllers can perform better than the expert in the experiments. In the industrial case study, comparisons between two DRL controllers using four and five observations indicate that constructing a comprehensive state is beneficial for training DRL controllers. Besides, the obtained standalone DRL

controller and the DRL-DRL cascade controller are slightly better than the PID-DRL cascade controller from our previous work. In general, the proposed approach improves the reliability and applicability of DRL controllers and provides an effective training method for reinforcement learning controllers to overcome large time delay issues.

CRediT authorship contribution statement

Zhoupeng Shou: Writing – original draft, Investigation, Data curation. **Xiaodong Hong:** Validation, Conceptualization. **Jingyuan Sun:** Data curation. **Yao Yang:** Methodology. **Jingdai Wang:** Project administration, Funding acquisition. **Yongrong Yang:** Resources, Formal analysis. **Zuwei Liao:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Zuwei Liao reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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